Group 4 Term Project - Pulsatile Backward Facing Step

Agniva Ghosh (Roll No: 25M1663)
Devraj (Roll No: 25D0871)
Manoj Kumar Kilana (Roll No: 25M1646)
Venkata Manoja Surya Veluri (Roll No: 25M1624)

Department of Mechanical Engineering, Indian Institute of Technology Bombay

Problem Statement

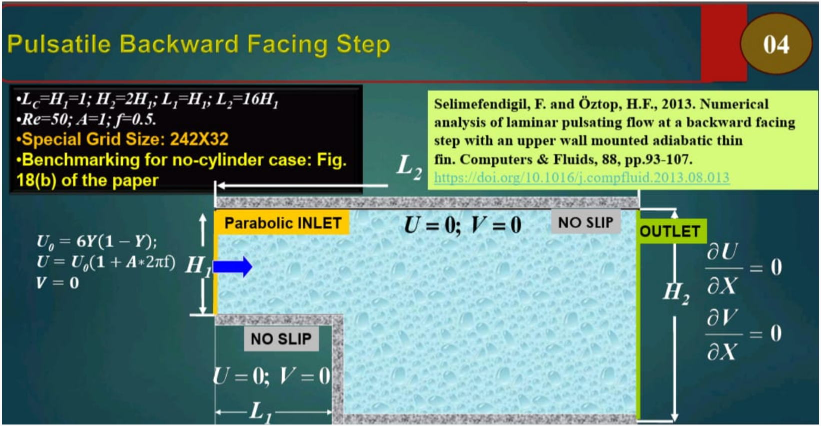


Figure 1: Problem Statement (Re=10 and 100 for Benchmark problem)

Governing Equations & Boundary Conditions

Governing Equations:

$$\frac{\partial U}{\partial X} + \frac{\partial V}{\partial Y} = 0 \tag{1}$$

$$\frac{\partial U}{\partial \tau} + U\frac{\partial U}{\partial X} + V\frac{\partial U}{\partial Y} = -\frac{\partial P}{\partial X} + \frac{1}{Re}\left(\frac{\partial^2 U}{\partial X^2} + \frac{\partial^2 U}{\partial Y^2}\right) \tag{2}$$

$$\frac{\partial V}{\partial \tau} + U\frac{\partial V}{\partial X} + V\frac{\partial V}{\partial Y} = -\frac{\partial P}{\partial Y} + \frac{1}{Re}\left(\frac{\partial^2 V}{\partial X^2} + \frac{\partial^2 V}{\partial Y^2}\right) \tag{3}$$

Boundary Conditions:

- **At Channel Inlet:** $U = 6Y(1 - Y) \cdot (1 + A \cdot sin(2\pi St\,\tau)), V = 0$
- **At Channel Exit:** $\frac{\partial U}{\partial X} = 0$, $\frac{\partial V}{\partial X} = 0$
- **On All other Channel Walls:** $U = 0$, $V = 0$
- **Pressure:** Inlet - $\partial P/\partial X = 0$, Outlet - $P = 0$, Horizontal Surface - $\partial P/\partial Y = 0$, Vertical Surface - $\partial P/\partial X = 0$
- Same BC for Pressure Correction as in Pressure

Methodology

- **Grid:** Staggered grid
- **Approach:** Finite Volume Method (FVM) with Flux based Pressure correction method
- **Advection scheme:** First Order Upwind (FOU)
- **Diffusion Term:** Central Difference (CD)
- **Grid Size:** 242 x 32
- **Changes from Steady to Unsteady in code:** (1) Outer while loop exit criteria changed w.r.t. iteration number (2) Unsteady BC implemented in U, V, U^*, V^*

Solution Algorithm

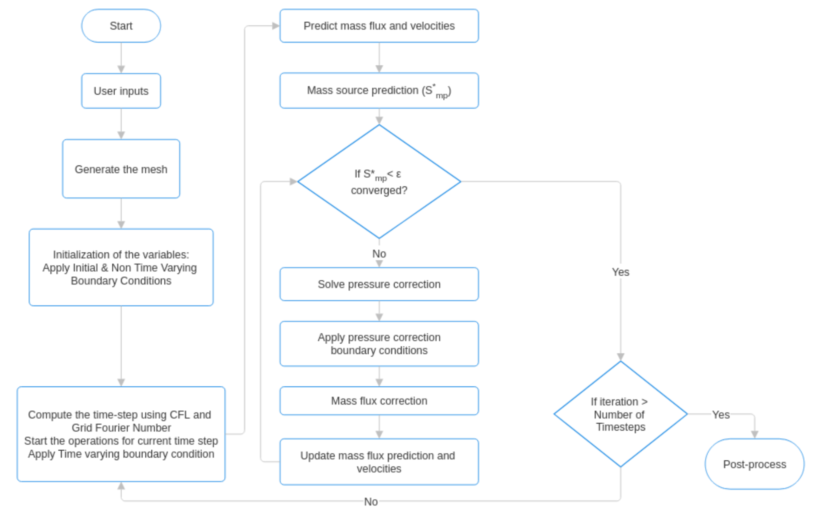
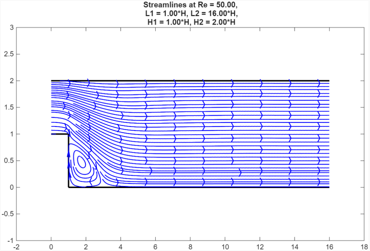
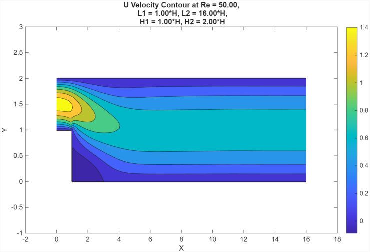


Figure 2: Computational Flowchart

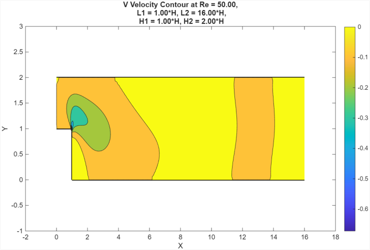
Results: Steady State Results ($L_2 = 16H$) (Inlet $U = 6Y(1-Y)$)



(a) Streamlines



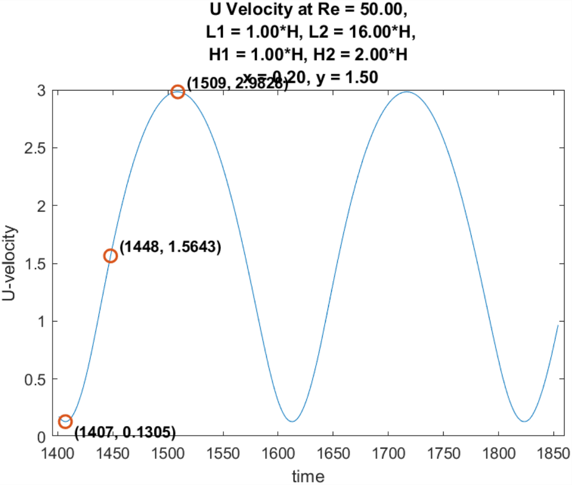
(b) U Velocity



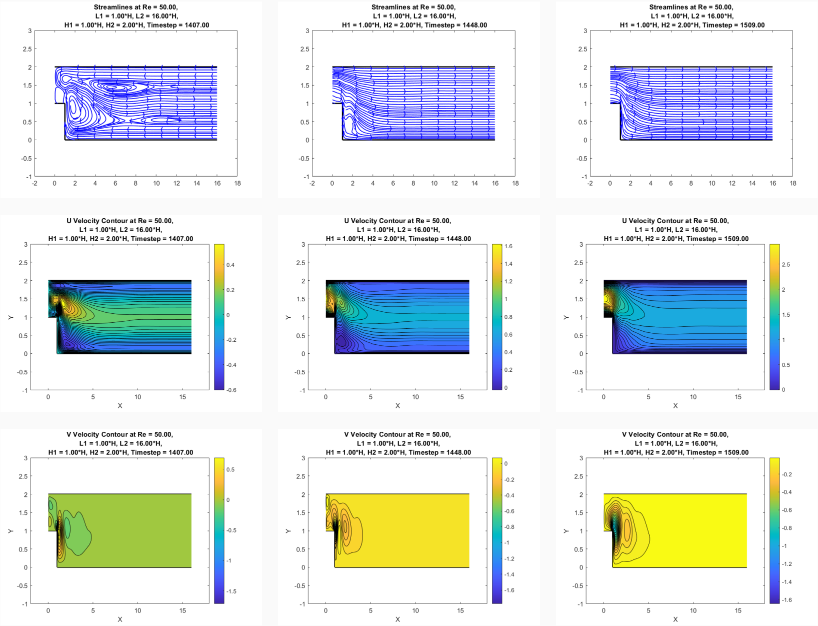
(c) V Velocity

Results: Original Case - Unsteady Results ($L_2 = 16H$, Re=50)

Time-steps where the maximum, mean and minimum U are recorded

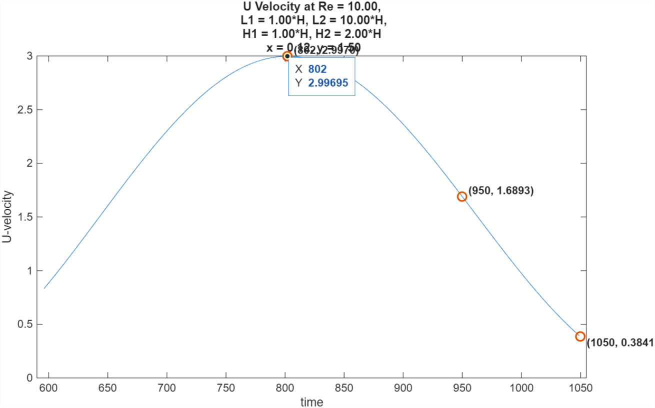


Results: Original Case - Unsteady Results ($L_2 = 16H$, Re=50)

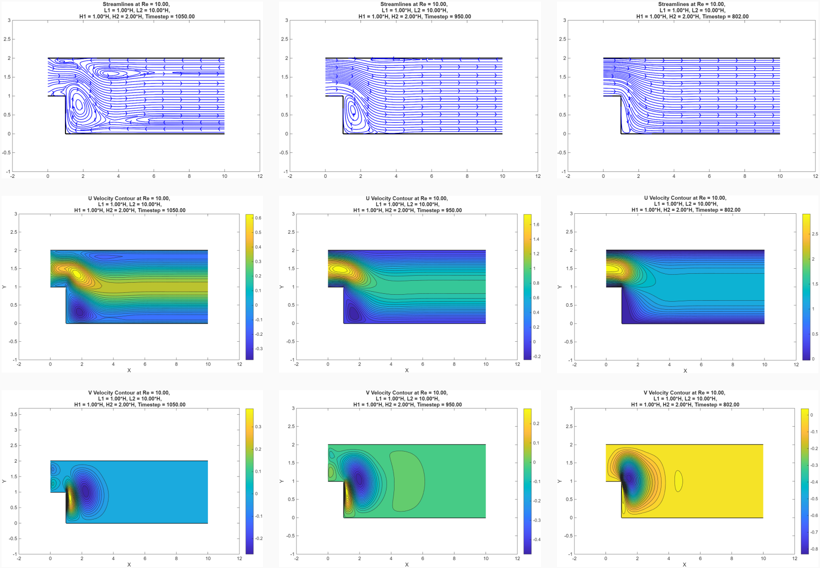


Results: Unsteady State Results ($L_2 = 10H$) (Re = 10)

Time-steps where the maximum, mean and minimum U are recorded

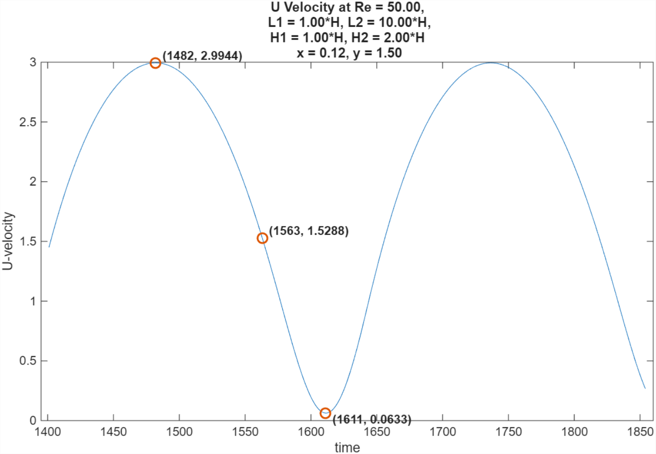


Results: Unsteady State Results ($L_2 = 10H$) (Re = 10)

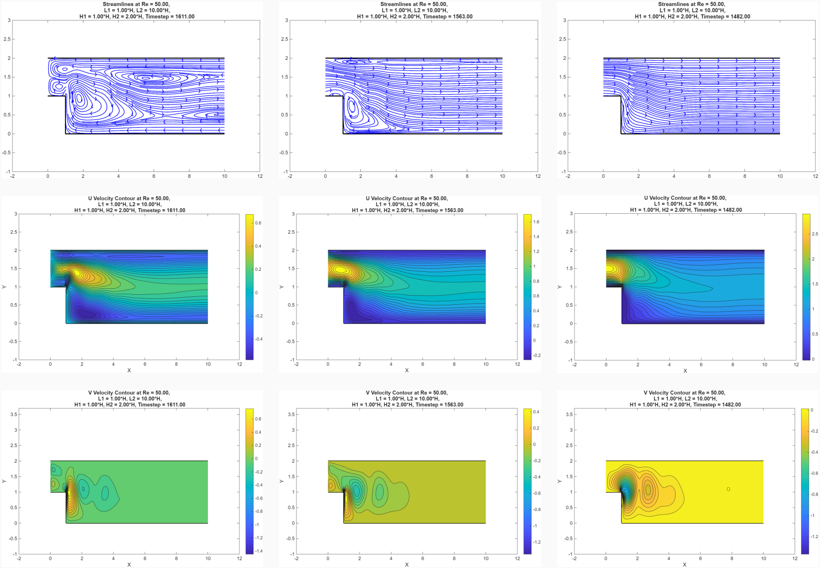


Results: Unsteady State Results ($L_2 = 10H$) (Re = 50)

Time-steps where the maximum, mean and minimum U are recorded

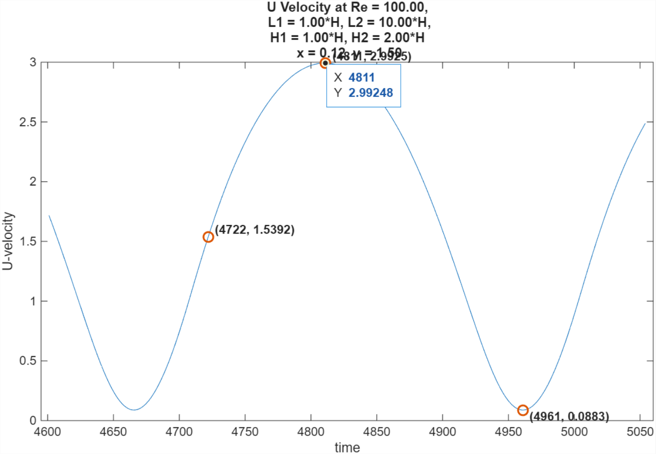


Results: Unsteady State Results ($L_2 = 10H$) (Re = 50)

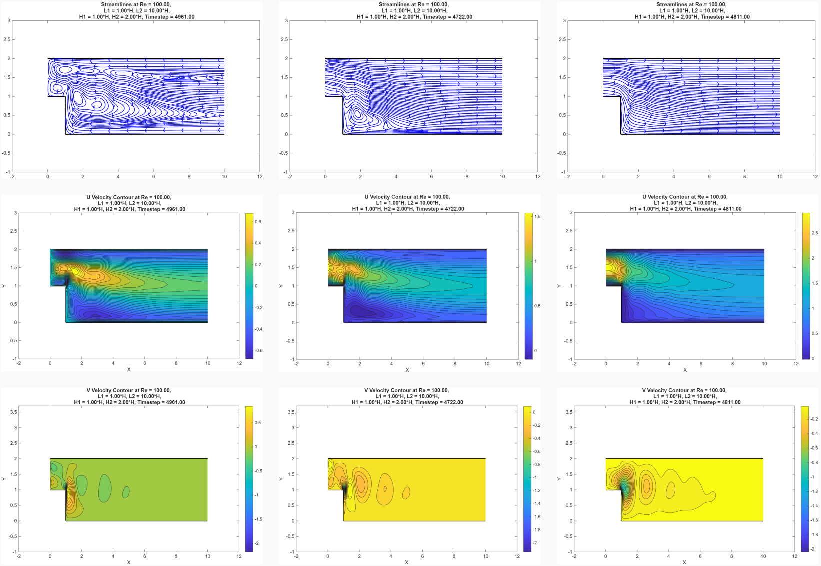


Results: Unsteady State Results ($L_2 = 10H$) (Re = 100)

Time-steps where the maximum, mean and minimum U are recorded

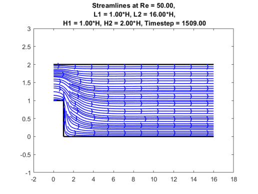


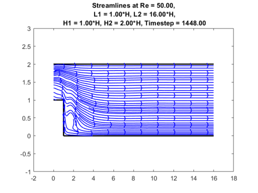
Results: Unsteady State Results ($L_2 = 10H$) (Re = 100)

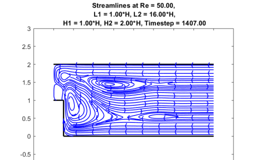


Mesh Independence

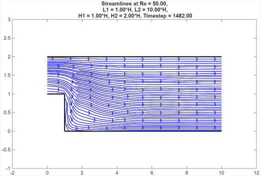
$\mathrm{Re} = 50,\ L_2 = 16H$

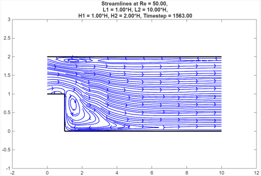


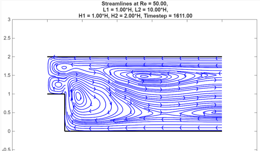




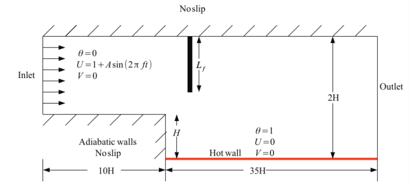
$\mathrm{Re} = 50,\ L_2 = 10H$



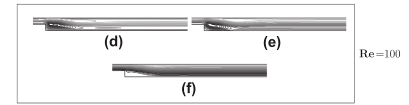




Benchmarking Results from Selimefendigil et al.



(a) Geometry



(b) Results

The timesteps that we have taken for the periodic variation study has been on the basis of U-velocity variation instead of Nusselt number variation in the paper. ($L_c = 2H$ in paper, $L_c = H$ for us. Hence our $Re = 50$ should be similar to paper's $Re = 100$ result. Fin and hot wall are not for our case)

Conclusions

- Unsteady backward facing step has been solved and results have been validated for the $L_2 = 16H$ case.
- Major Trend - For all Re values, at maximum inlet velocity instance, there is no vortex observed in the step; at the average inlet velocity instance there is presence of some vortices in the flow, and for the minimum inlet velocity the most vortices are observed in the flow.
- When during pulsation, when velocity is low, viscous forces are comparable to inertia forces and this slows down the fluid, causing circulation in some regions. When velocity is high inertia forces dominate over viscous force and there is no circulation observed.
- Results for Re 10, 50 and 100 have been shown for the shortened $10H$ domain
- Mesh independence has been achieved for given grid size, by repeating for $16H$ and $10H$ for the same grid points, which show a high degree of resemblence